

Founding-cohort species classes, and the complexity of IAP

Resolving Conjecture S1 (note 06) for infinite clusters, and Conjecture 2b (note 09)

27 September 2026

Provenance. Produced by OpenAI Codex during an external audit of this repository (27 September 2026), and checked by Anthropic Claude before inclusion. Hand proofs; no independent human review; priority not established.

Source. S. A. Alexander, *Specieslike clusters based on identical ancestor points*, J. Mathematical Biology, 2026 ([arXiv:2602.05274](https://arxiv.org/abs/2602.05274)): Definitions 1–6 and 8–10, and Theorems 6, 8, 9 and 13. The founding-cohort construction is note 06 of this repository.

Results.

- *Theorem 1.* Infinite maximal founding-cohort clusters and infinite maximal common-ancestor clusters represent exactly the same $\sim$ -classes. This proves Conjecture S1, restricted to infinite sets as the source’s definition of $\sim$ requires.
- *Theorem 2.* Global IAP is Π^0_3 -complete, even on connected rooted binary trees. This proves Conjecture 2b of note 09.

1. Founding-cohort and common-ancestor species give the same infinite classes

Write

$$\mathcal{C} = \text{IAP} \cap \text{CONV} \cap \text{CA} \cap \text{REF}, \quad \mathcal{F} = \text{CONN} \cap \text{IAP} \cap \text{CONV} \cap \text{REF} \cap \text{FC}.$$

Here IAP prohibits any member from having both infinitely many descendants and infinitely many non-descendants in the set. CONV includes a vertex whenever it has an ancestor and a descendant in the set. REF requires any included vertex with infinitely many descendants in the biosphere to have infinitely many in the set. CA requires an included common ancestor; FC requires all internal founders to precede all internal non-founders. Connectedness is undirected.

For an infinite set S , $\text{gen}(S)$ consists of members having only finitely many non-descendants in S . Alexander’s equivalence $S \sim T$ means $\text{gen}(S) \Delta \text{gen}(T)$ is finite; **the published definition is restricted to infinite sets**. The unqualified conjecture needs that restriction.

Theorem 1. Every infinite inclusion-maximal member of $\mathcal{F}$ is $\sim$ -equivalent to an infinite inclusion-maximal member of $\mathcal{C}$, and conversely. Thus the sets of represented $\sim$ -classes agree.

Proof, forward direction. Let S be an infinite maximal member of $\mathcal{F}$. It has a non-founder: a connected set without any internal edge is a singleton. All founders precede any fixed non-founder, so there are only finitely many founders. Following internal parents terminates, since

only finitely many vertices precede any fixed birth. Every vertex is therefore a founder or a descendant of a founder. At least one founder r has infinitely many descendants in S .

By IAP at r , the set

$$S_r = \{r\} \cup (\text{desc}_G(r) \cap S)$$

is cofinite in S . It belongs to $\mathcal{C}$:

- r is its common ancestor.
- IAP is inherited by subsets.
- Convexity follows from convexity of S and transitivity of ancestry.
- If $x \in S_r$ has infinitely many descendants in G , REF for S supplies infinitely many in S . All of these descend from r and hence lie in S_r . Thus REF holds.

By Alexander's upward-genericity theorem and Zorn's lemma, extend S_r to an inclusion-maximal $M \in \mathcal{C}$. It is infinite. Finite branching and convexity give an infinite directed path starting at r within S_r . Every vertex on this path has infinitely many descendants in both S and M . IAP makes every one of them a generator of both sets. Consequently

$$|\text{gen}(S) \cap \text{gen}(M)| = \infty.$$

Both S and M are infinite specieslike clusters. Alexander's Objective Species Theorem (Theorem 6) now gives $S \sim M$. This is the missing comparison in S1.

Reverse direction. Let M be an infinite maximal member of $\mathcal{C}$. CA plus CONV implies CONN and FC: the common ancestor is the only internal founder. Thus $M \in \mathcal{F}$, and the founding-cohort upward-genericity theorem extends M to a maximal $S \in \mathcal{F}$. An infinite path in M again supplies infinitely many common generators of M and S . The same Objective Species Theorem gives $M \sim S$. ■

The proof does not show that S and M themselves have finite symmetric difference, that maximal representatives are unique, or that all representatives are comparable by inclusion. The forward direction only needs maximality to match the conjecture's requested output; the argument works for every infinite $S \in \mathcal{F}$. Finite maximal clusters can exist, but $\sim$ is not defined for them in the cited paper.

2. Global IAP is Π^0_3 -complete, even for rooted binary trees

We use a precise observation space. Vertices are numbered by birth order $0, 1, 2, \dots$, with birth-date $t(n)=n$. The n th coordinate specifies the finite subset of $\{0, \dots, n-1\}$ that parents vertex n . Each coordinate has the discrete topology; observations reveal initial coordinates. Ancestry of a fixed earlier vertex to vertex n is decided by the first $n+1$ coordinates.

Let $\mathcal{B}$ be the subspace in which vertex 0 is the only root, every other vertex has exactly one parent, and every vertex has at most two children. This is a closed subspace of the coordinate product: violations have finite witnesses. Its members are connected infinite biospheres satisfying all four conditions of Alexander's Definition 1.

Theorem 2. The set of members of $\mathcal{B}$ satisfying global IAP is Π^0_3 -complete under continuous reductions from Cantor space. Global IAP on the larger birth-coded biosphere space has the same relative upper bound and is at least this hard.

Upper bound. Write $D(v, n)$ for strict ancestry. IAP says precisely

$$\forall v \exists (N, b) \in \mathbb{N} \times \{0, 1\} \forall n \geq N [D(v, n) = b].$$

The bracketed predicate is clopen. This is a Π^0_3 definition. It asserts eventual constancy separately for each vertex, not one common cutoff for all.

Hardness. Reduce

$$P_3 = \{x \in 2^{\mathbb{N} \times \mathbb{N}} : \forall k \{j : x(k, j) = 1\} \text{ is finite}\}.$$

Construct a permanent backbone $b_0 \rightarrow b_1 \rightarrow b_2 \rightarrow \dots$. At b_k attach a side founder a_k . For each 1 in row k , append one new child to the current tip of a_k 's side chain. Zeros add nothing to that chain. Never join a side chain back to the backbone or to another side chain.

To make births and continuity explicit, at stage s :

1. Create b_s , with parent b_{s-1} when $s > 0$.
2. Create a_s , with parent b_s .
3. For $k=0, \dots, s$ in order, inspect $x(k, s-k)$; if it is 1, extend side chain k by one vertex.

Assign the created vertices consecutive birth numbers. The backbone supplies infinitely many births. Every parent is already born; each backbone vertex has exactly two children, and every side-chain vertex has at most one. Thus $G_x \in \mathcal{B}$. The first m output vertices depend on only finitely many input bits (stages through m suffice), so $x \mapsto G_x$ is continuous.

If some row k has infinitely many 1s, a_k has infinitely many descendants along its side chain and infinitely many non-descendants along the backbone. IAP fails at a_k .

If every row has finitely many 1s, every side-chain vertex has only finitely many descendants. For b_k , its non-descendants lie among the finitely many earlier backbone vertices and the **finitely many finite** side chains with indices below k , plus b_k itself. Every later backbone vertex and every side chain with index at least k descends from b_k . Thus b_k has only finitely many non-descendants. Every vertex satisfies IAP.

Therefore

$$G_x \in \text{IAP} \iff x \in P_3.$$

This answers this repository's concern about unwanted interaction between rows: put rows successively along a backbone. Only finitely many earlier rows can contribute non-descendants to any one backbone vertex. ■

Why P_3 is complete (included to make the reduction self-contained). Write any Π^0_3 subset A of Cantor space as $\bigcap_k \bigcup_N F_{k,N}$, where the F 's are closed and increasing in N . At stage s let $m_k(s)$ be the least $N \leq s$ for which the cylinder of $x \upharpoonright s$ meets $F_{k,N}$, or $s+1$ if none does. For fixed k this integer sequence is nondecreasing. It is bounded iff $x \in \bigcup_N F_{k,N}$: a surviving closed candidate eventually bounds it, whereas every fixed candidate is eventually excluded when x belongs to none. Output a 1 in row k exactly when $m_k(s)$ increases from its preceding value, starting with $m_k(-1)=0$. Each output bit depends on a finite input prefix. That row is finite iff the integer sequence is bounded, giving a continuous reduction $A \leq P_3$. The immediate $\forall k \exists N \forall j \geq N$ zero-tail definition supplies P_3 's upper bound. This is topological completeness; no effectiveness of closed set presentations is silently assumed.

Corollary. Failure of uniform eventual learning persists on connected, single-founder, outdegree-two biospheres. Moreover, even the fixed side founder a_0 satisfies IAP iff its row has finitely many 1s. Taking all other rows zero reduces FIN to that single-organism predicate. FIN is a countable dense F_σ subset of Cantor space and not G_δ , hence is not guessable even by a noncomputable finite-prefix guesser. This strengthens this repository's disconnected two-lineage example without making a claim about learning every possible species-valued output.

Status and research connection

The first theorem is a short consequence of Alexander's existing objective species theorem plus this repository's founding-cohort result. The second supplies an explicit reduction for this repository's proposed complexity classification. On locally finite rooted trees it is closely related to uniqueness of an infinite end; that connection is a reason to be cautious about external novelty, not a gap in the reduction.

No Lean check was run. The audit controls (`../audit-2026-09-27/controls.py`) check finite instances of the construction's parent, degree, prefix, and descendant invariants; the infinite conclusions rest on the proofs above.